

Full Report For Student Management Project

Comprehensive Academic Data System (ADS) Architecture Report

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Target Platform: PostgreSQL Engine v15+

1. Executive Summary

This project outlines the engineering, seeding, auditing, and optimization of a scalable educational relational database model designed to handle 16 linked entities. The architecture transitions from an Operational Data Store (ODS) handling users, schedules, and financial records into an analytical reporting framework suited for Business Intelligence (BI) integration.

2. Structural Architecture & Relationship Modality

The schema uses custom UUID identifiers (``uuid_generate_v4()``) to ensure absolute record uniqueness across distributed nodes.

Core Relationship Mechanics:

* **One-to-One Sub-typing:** Primary identities map uniquely from the ``users`` base table into individual roles (``students``, ``teachers``, ``parents``).

* **Circular Link Resolution:** To prevent relational deadlocks during row synchronization, cross-table constraints on ``users`` pointing back to sub-types were dropped. The system safely enforces strict downward data integrity (Sub-types reference ``users.user_id``).

* **Operational Junctions:** Core activities use intermediate transactional schemas (e.g., ``enrollments`` links students to classes, ``student_parents`` registers multi-guardian contact rules).

3. Performance Optimization Strategy

To counter the high disk fragmentation risk inherent to random UUID structures, explicit secondary Indexes were deployed across all frequently queried Foreign Keys:

* ``idx_attendance_enrollment`` & ``idx_attendance_schedule`` optimize daily check logs.

* ``idx_submissions_assessment`` guarantees fast lookup for grading metrics.

* ``idx_fees_student`` & ``idx_payments_fee`` avoid full-table scans during financial lookups.

4. Analytical Transformation

A master database View (``analytics_student_master_dashboard``) was built to abstract multi-table joins. This view flattens granular rows into unified records calculating:

1. Student demographic profiles
2. Attendance rates (Percentage calculation)
3. Normalized assignment scores
4. Net outstanding accounting financial balances

This view allows BI engines (Power BI/Tableau) and analytical runtimes (Python Pandas) to execute lightning-fast aggregations without joining raw source entities.